

SARAN AV

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PROFESSIONAL SUMMARY

Final-year B.E. Electrical & Electronics Engineering student at Panimalar Engineering College (CGPA 8.32) with strong foundations in Java programming and basics in C programming, front-end technologies including HTML and CSS. Hands-on experience in developing academic and personal projects with a passion for software development and my core field projects. Additionally, I had knowledge in Advanced concepts in power electronics. Built and deployed end-to-end projects including an AI-based water quality anomaly detection system and a face-recognition attendance platform. Seeking an opportunity to apply my skills and contribute to the organizational growth, enhance my expertise in challenging and dynamic environment.

TECHNICAL SKILLS

Programming Languages: Java, C

Embedded & IoT: ESP32, Arduino, PLC, Sensor Integration, Real-time Dashboards

Web Technologies: Front-end (HTML, CSS, Basics in Javascript)

Tools & Platforms: MATLAB/Simulink, Git, GitHub Pages, HuggingFace Embeddings

Concepts: OOP, Data Structures & Algorithms, DBMS, Power Systems

Soft Skills: Leadership, analytical thinking, problem-solving, technical communication, teamwork

PROJECTS

AI-Based Water Quality Anomaly Detection System 2025

ESP32 · Python · Z-Score Statistical Methods · IoT Sensors · OLED Display

- Built an ESP32-based system using a turbidity sensor and Z-Score statistical methods to detect anomalies in water quality data in real time.
- Integrated IoT sensor data pipelines with a Python-based model and a real-time WiFi dashboard for live monitoring.
- Authored a 36-page project report and interactive HTML dashboards; applicable to smart water management and environmental monitoring.

Face Recognition Smart Attendance Management System 2026

face-api.js · HTML/CSS/JavaScript · GitHub Pages

- Developed a single-file, browser-based attendance system using face-api.js for real-time face detection and recognition.
- Implemented client-side data persistence with localStorage, requiring no backend server.
- Deployed as a static app on GitHub Pages, showcasing practical applied-AI and frontend engineering skills.

PDF Q&A System using RAG Architecture 2026 (Amazon Summer School)

Python · Streamlit · LangChain · Gemini API · FAISS · HuggingFace Embeddings

- Built a Retrieval-Augmented Generation pipeline enabling natural-language question answering over uploaded PDF documents.
- Used FAISS for vector similarity search and HuggingFace embeddings for semantic document indexing.
- Deployed an interactive Streamlit interface and a companion portfolio website for the project.

EDUCATION

B.E. Electrical and Electronics Engineering | Panimalar Engineering College Sept 2023 – Present

CGPA: 8.30 (upto 6th Sem)

Higher Secondary (HSC) | St. John's Matriculation Higher Secondary School, Porur, Chennai Jun 2022 – Apr 2023

489/600 – 81.5%

SSLC | St. John's Matriculation Higher Secondary School, Porur, Chennai Jun 2020-Apr 2021

ALL PASS

CERTIFICATIONS

- Frontend Developer – SkillHub
- Advanced Power Electronics – NPTEL

INTERNSHIP

Intern | Priya Plastic, Chennai2025

- Gained hands-on exposure to industrial operations, plastic raw material supply chain, and manufacturing processes.